Exception contracts and guided repairs

Research release report | 16 October 2026

Original finding remains unchanged

An ItsDangerous timestamp repair passed 297 upstream tests while replacing the original bytes payload with diagnostic text. Two feedback repairs did not fix that exception-contract failure. Later repairs in other repositories do not resolve this original case.

Executable contract audit

ECA-6 is six related templated tasks with 36 constructed faulty variants. The weak gate accepted all 36. The first contract gate rejected 35/36; six model repairs passed. A post-outcome v2 gate rejected 36/36 mutants, accepted 5/6 model repairs and accepted 6/6 references. The stronger gate was designed after outcomes and is not independent validation. Local replay reproduces 102 variants and compares every recorded case and semantic assertion.

Additional repositories: final assisted patches

Case	Upstream	Examples	Withheld	Final total
Tenacity: falsey wait callable	142	8	9	159/159
Packaging: pylock filename boundary	74	7	9	90/90

Both final patches pass 249 checks in total. Tenacity repairs synchronous and asynchronous wait handling. Packaging preserves the valid base filename and rejects invalid named filenames. A post hoc comparison against the pinned fixed Packaging function found zero differences across 1,245 names; this is additional regression evidence, not a fresh withheld benchmark.

Assistance, failures and scope

These are adaptive repair case studies. Initial and feedback attempts failed. The final Tenacity patch used staged diagnosis and literal-function output. Packaging required explicit guidance to separate two valid filename families into branches after earlier repairs still failed. Public commit descriptions and known source localization were provided. This is not an autonomous or contamination-free first-attempt success rate.

The final generated function bodies were applied without manual semantic edits, using recorded source replacement rules and retained surrounding decorators. One Tenacity diagnosis incorrectly suggested that 0 versus 0.0 itself explained the failure; passing tests do not endorse that explanation. Failed outputs and preflight corrections remain available in the evidence archive.

Model and reproducibility

Local model: Gemma 4 E2B QAT, Q4_0 GGUF; revision 675cff42a74c774d6cb76f76d8eacb49b48c9b93. Runtime: llama.cpp b11429, commit d81235049. Recorded settings: context 4096, temperature 0, seed 20261016, two CPU threads, reasoning off. Model and prompt digests are checked by the release tools.

reproduce.py replays ECA without a model. replay_repairs.py verifies pinned upstream archive hashes and reruns fixed, buggy-parent and final model variants. generate_repairs.py --plan lists the final guided prompts and settings; fresh generation records new outputs separately and does not certify them. FETCH_UPSTREAM.md supplies exact revisions and archive digests.

Evidence and publication boundary

FOLLOWUP.md separates these findings from the unchanged original submitted paper. outputs/ retains protocols, prompts, model responses, failures, reviewed patches and results. Original tooling is Apache-2.0; third-party licenses and notices are preserved. The original Kaggle writeup remains under its accepted terms. Model weights, runtime binaries, complete upstream archives and account records are excluded. This package does not publish or submit an update.